



Specification Hardcopy Managed Print

Document Number:	S_API_3629
Document Title:	Spec_API_01-1548
Effective Date:	01 Oct 2024 09:48:27 (GMT-04:00)
Document Signatories	

Signed By : Bhargav Patel (bhargavp)

Decision : Approved

Decision Date : 26 Sep 2024 17:41:45 (GMT-04:00)

Role : Approver

Purpose : Revise the Specification

Meaning Of Signature : I have checked this document and approve it for use

Signed By : Manaliben K. Patel (manalibenk)

Decision : Approved

Decision Date : 27 Sep 2024 14:45:52 (GMT-04:00)

Role : QA Authoriser

Purpose : Revise the Specification

Meaning Of Signature : I am granting the Quality Authorisation for the use of this document

Signed By : Gira Shah (gira.shah)

Decision : Approved

Decision Date : 01 Oct 2024 09:48:05 (GMT-04:00)

Role : Document Control Authorizer

Purpose : Revise the Specification

Meaning Of Signature : I am granting the Document Control Authorization for the use of this document.

Print Information	
This is a Hardcopy Managed Print for GMP use. A record has been made of this print. An automatic message will be sent when this document changes.	
Printed By:	Nilpa Patel (nilpa.patel)

RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM				
Material Name: Estradiol, USP (Hemihydrate, Micronized) (Material manufactured by: Aspen Oss B.V.)				
Code No.: 01-1548	Specification No.: 01-1548-04	Page 1 of 5		
Sample Information				
Manufacturer Lot No.:	Amneal Receiving No.:	Laboratory Receiving No.:		
Sample Type: ☐ FT (Full testing) ☐ RA (Re-assay) ☐ ERT (Enhanced Reduced Testing)				
¹ Test/Method No.	RA	ERT	Specification	Results
Description Visual examination GM-DES	X	X	White to practically white powder.	
				Performed By/ Date:
				Book/Pg. #
Solubility		X	Soluble in alcohol, in acetone, in dioxane and in solutions of fixed alkali hydroxides; slightly soluble in chloroform; sparingly soluble in vegetable oils; practically insoluble in water.	
				Performed By/ Date:
				Book/Pg. #
Identification A <197M>	X	X	The Infrared absorption spectrum of the sample exhibits maxima at the same wavelengths as that of similar preparation of standard.	
				Performed By/ Date:
				Book/Pg. #
⁷ Identification B <197U>			Absorptivities at 280 nm, calculated on the anhydrous basis, should not differ by more than 3.0%.	
				Performed By/ Date:
				Book/Pg. #

RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM				
Material Name: Estradiol, USP (Hemihydrate, Micronized) (Material manufactured by: Aspen Oss B.V.)				
Code No.: 01-1548	Specification No.: 01-1548-04	Page 2 of 5		
Sample Information				
Amneal Receiving No.:			Laboratory Receiving No.:	
¹Test/Method No.	RA	ERT	Specification	Results
⁷ Melting Range <i>Class I <741></i>			Between 173°C and 179°C.	
				Performed By/ Date:
				Book/Pg. #
⁷ Specific Rotation <i><781S></i>			Between +76° and +83° at 25°C.	
				Performed By/ Date:
				Book/Pg. #
^{6, 7} Water <i><921>, Method I</i>	X		NMT 3.5%.	
				Performed By/ Date:
				Book/Pg. #
^{2, 7} Residual Solvents GM-RS			Ethanol : NMT 2000 ppm	
			Methanol : NMT 200 ppm	
				Performed By/ Date:
				Book/Pg. #

RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM				
Material Name: Estradiol, USP (Hemihydrate, Micronized) (Material manufactured by: Aspen Oss B.V.)				
Code No.: 01-1548	Specification No.: 01-1548-04	Page 3 of 5		
Sample Information				
Amneal Receiving No.:			Laboratory Receiving No.:	
¹Test/Method No.	RA	ERT	Specification	Results
⁷ Assay 01-1548-AS	X		Not less than 97.0% and not more than 103.0% of C ₁₈ H ₂₄ O ₂ , calculated on the anhydrous basis. Report assay on as is basis.	Anhydrous basis:
				As is basis:
				Performed By/ Date:
				Book/Pg. #
^{3, 7} Related Compounds 01-1548-RC	X		Specified Identified Impurities:	
			Δ ^{9,11} - Estradiol : NMT 0.1%	
			17α-Estradiol : NMT 0.25%	
			Estrone : NMT 0.2%	
			Any Unspecified Impurity: NMT 0.10%	
			Total Impurities : NMT 0.5%.	
				Performed By/ Date:
	Book/Pg. #			
^{4, 7} Particle Size 01-1548-PS			D ₉₀ : NMT 5 μm.	
			D ₉₉ : NMT 15 μm.	
				Performed By/ Date:
				Book/Pg. #

RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM				
Material Name: Estradiol, USP (Hemihydrate, Micronized) (Material manufactured by: Aspen Oss B.V.)				
Code No.: 01-1548	Specification No.: 01-1548-04	Page 4 of 5		
Sample Information				
Amneal Receiving No.:			Laboratory Receiving No.:	
¹Test/Method No.	RA	ERT	Specification	Results
⁷ Visual Impurities			Not present.	
				Performed By/ Date:
				Book/Pg. #
Vendor COA		X	The test results reported on the material Certificate of Analysis (COA) from the manufacturer/vendor, meet the manufacturer/ vendor specifications.	
Re-test Date:			Expiration Date:	
Remarks/Comments:				

Foot Notes:

1. All tests to be performed using the current USP monograph of Estradiol except where noted for in-house method.
2. Per the Manufacturer's (Aspen Oss B.V) DMF, Class 2 and Class 3 solvent is likely to be present. None of the Class 1 solvents as such are used in the manufacturing process of Estradiol, USP (Hemihydrate, Micronized).
3. Impurity's IUPAC name:

Vendor's Name	EP Name	IUPAC Name
$\Delta^{9,11}$ -Estradiol	Impurity D	estra-1,3,5(10),9(11)-tetraene-3,17 β -diol
17 α -Estradiol	Impurity B	estra-1,3,5(10)-triene-3,17 α -diol
Estrone	Impurity A	3-hydroxyestra-1,3,5(10)-trien-17-one

4. Additional in-house test control consistent with vendor's COA.
5. Abbreviations: NA: Not applicable, NLT: Not less than, NMT: Not more than, X: Tests to be performed, ND: Not Detected, LOQ: Limit of Quantification.
6. No alert notification or alert investigation is required for water content, if obtained result is within 1.0% of vendor reported result.



RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM

Material Name: Estradiol, USP (Hemihydrate, Micronized)
(Material manufactured by: Aspen Oss B.V.)

Code No.: 01-1548

Specification No.: 01-1548-04

Page 5 of 5

Sample Information

Amneal Receiving No.:

Laboratory Receiving No.:

7. Vendor reported result will be reported on release COA for the test which are not performed as part of Enhance Reduced Testing.
8. One lot of each material from a qualified vendor shall be fully tested every ten lot or 3 years whichever is earlier according to approved specifications. (As per current SOP P_CQCM_0232).

Material Certification

Reviewed By: _____
Laboratory

Date: _____

☐ Approved ☐ Rejected By: _____
Analytical/Quality Control Laboratory

Date: _____

RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM	
Material Name: Estradiol, USP (Hemihydrate, Micronized) (Material manufactured by: Aspen Oss B.V.)	
Code No.: 01-1548	Specification No.: 01-1548-04

Document History

Version No.	CC #	Changes	Reason for Changes
01	CC_2018_1997	This specification is adopted from the previously approved specification number 1381-02, effective date: 02 June 2017.	New code generated for commercial manufacturing of Estradiol transdermal systems.
02	CC_2019_2012	Added GM-DES method reference to the description test.	To use GM-DES method to perform Description test.
		Reworded the specification of Solubility test.	To comply with current USP.
03	CC-AMN-22-2002	Included foot note # 6 for water content test "No alert notification or alert investigation is required for water content, if obtained result is within 1.0% of vendor reported result."	Based on the historical data, vendor results of water content for estradiol USP is within specification limit and always beyond the alert limit as per Amneal alert SOP P_CQCM_0233. Hence no further action is required to initiate the alert notification.
		Removed date from the header	As per current documentation practice
04	CC-AMN-24-4869	-Updated the heading to Enhanced Reduced Testing (ERT) from Reduced testing (RT). -For Test Description, Identification A, Identification B, Melting Range, Specific Rotation, Water, Assay, Related Compounds, Particle Size, Visual Impurities and Vendor COA are removed from RT conditions and Description, Solubility, Identification A, and vendor COA added in ERT also added superscript no.in respective test.	-As per Enhanced reduced testing report (Report No: PRT_MP_QC_PW_373_23)



RAW MATERIAL SPECIFICATIONS / LABORATORY TEST REPORT FORM	
Material Name: Estradiol, USP (Hemihydrate, Micronized) (Material manufactured by: Aspen Oss B.V.)	
Code No.: 01-1548	Specification No.: 01-1548-04

Version No.	CC #	Changes	Reason for Changes
		-Incorporated Footnote#7 as “Vendor reported result will be reported on release COA for the test which are not performed as part of Enhance Reduced Testing”. -Incorporated footnote#8 “One lot of each material from a qualified vendor shall be fully tested every ten lot or 3 years whichever is earlier according to approved specifications. (As per current SOP P_CQCM_0232).	-Amneal will perform testing as per SOP# P_CQCM_0232 with the proposed enhanced reduced testing program.

Printed on: Wed Jan 28 2026, 10:57:35 am

Printed by: Gaurav Rami

Status: Currently Official on 28-Jan-2026

Official Date: Official as of 01-May-2020

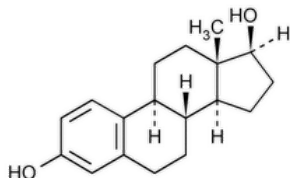
Document Type: USP Monographs

DOI: https://doi.org/10.31003/USPNF.M30610_02_01Printed from: https://online.uspnf.com/uspnf/document/1_GUID-9913954A-8B7D-4C49-9A09-955EF6ED9C76_2_en-US

© 2026 USPC

Do not distribute

Estradiol

C₁₈H₂₄O₂ 272.38

Estra-1,3,5(10)-triene-3,17-diol, (17β)-.

Estra-1,3,5(10)-triene-3,17β-diol CAS RN®: 50-28-2; UNII: 4TI98Z838E.

Hemihydrate 281.39 CAS RN®: 35380-71-3; UNII: CXY7B3Q98Z.

» Estradiol contains not less than 97.0 percent and not more than 103.0 percent of C₁₈H₂₄O₂, calculated on the anhydrous basis.**Packaging and storage**—Preserve in tight, light-resistant containers. Store at 25°, excursions permitted between 15° and 30°.**Labeling**—The hemihydrate form is so labeled.**USP REFERENCE STANDARDS** (11)—[USP Estradiol RS](#)[USP Estrone RS](#)**Identification**—**Change to read:****A:** ▲ [SPECTROSCOPIC IDENTIFICATION TESTS \(197\), Infrared Spectroscopy: 197M](#) ▲ (CN 1-May-2020) ·**Change to read:****B:** ▲ [SPECTROSCOPIC IDENTIFICATION TESTS \(197\), Ultraviolet-Visible Spectroscopy: 197U](#) ▲ (CN 1-May-2020) —*Solution:* 50 µg per mL.*Medium:* alcohol.

Absorptivities at 280 nm, calculated on the anhydrous basis, do not differ by more than 3.0%.

MELTING RANGE, Class I (741): between 173° and 179°. [NOTE—Dry over silica gel for not less than 16 hours prior to testing.]**SPECIFIC ROTATION (781S):** between +76° and +83°.*Test solution:* 10 mg per mL, in dioxane.**WATER DETERMINATION, Method I (921):** not more than 3.5%.**Chromatographic purity**—[NOTE—Make all solutions fresh daily.]*Mobile phase*—Prepare a filtered and degassed mixture of 2,2,4-trimethylpentane, *n*-butyl chloride, and methanol (45:4:1). Make adjustments if necessary (see *System Suitability* under [Chromatography \(621\)](#)).*Diluting solution*—Prepare a filtered and degassed mixture of *n*-butyl chloride and methanol (5:1).*Test solution*—Transfer about 70 mg of Estradiol, accurately weighed, to a 10-mL volumetric flask, dissolve in *Diluting solution*, shake vigorously to aid dissolution, dilute with *Diluting solution* to volume, and mix.*Chromatographic system* (see [CHROMATOGRAPHY \(621\)](#))—The liquid chromatograph is equipped with a 280-nm detector and a 4.6-mm × 25-cm column that contains packing L3. The flow rate is about 2 mL per minute. Chromatograph the *Test solution*, and record the peak responses as directed for *Procedure*: the resolution, *R*, between estradiol and any impurity is not less than 1.0; the column efficiency is not less than 800 theoretical plates; the tailing factor is not more than 1.5; and the relative standard deviation for replicate injections is not more than 2.0%.*Procedure*—Inject a volume (about 10 µL) of the *Test solution* into the chromatograph, record the chromatogram, and measure the peak responses. Calculate the percentage of each impurity in the portion of Estradiol taken by the formula:

$$100(r_i/r_s)$$

in which r_i is the peak response for each impurity; and r_s is the sum of the responses of all the peaks: not more than 0.5% of any individual impurity is found; and not more than 1.0% of total impurities is found.

Assay—

Mobile phase—Prepare a filtered and degassed mixture of acetonitrile and water (55:45). Make adjustments if necessary (see *System Suitability* under [Chromatography \(621\)](#)).

Internal standard solution—Transfer about 300 mg of ethylparaben to a 500-mL volumetric flask, add methanol to volume, and mix.

Standard preparation—Dissolve accurately weighed quantities of [USP Estradiol RS](#) and [USP Estrone RS](#) in methanol to obtain a solution containing 0.40 mg and 0.24 mg, respectively, in each mL. Pipet 10 mL of this solution and 5 mL of the *Internal standard solution* into a 200-mL volumetric flask. Add 100 mL of methanol, dilute with water to volume, and mix to obtain a solution having a known concentration of about 20 µg of [USP Estradiol RS](#) per mL.

Assay preparation—Transfer about 100 mg of Estradiol, accurately weighed, to a 250-mL volumetric flask, add methanol to volume, and mix. Transfer 10.0 mL of this solution to a 200-mL volumetric flask, add 5.0 mL of *Internal standard solution* and 100 mL of methanol, dilute with water to volume, and mix.

Chromatographic system (see [CHROMATOGRAPHY \(621\)](#))—The liquid chromatograph is equipped with a 205-nm detector and a 3.9-mm × 30-cm column that contains packing L1. The flow rate is about 1 mL per minute. Chromatograph the *Standard preparation*, and record the peak responses as directed for *Procedure*: the relative retention times are about 0.7 for the internal standard, about 1.3 for estrone, and 1.0 for estradiol; the resolution, R , between the analyte and estrone is not less than 2.0; and the relative standard deviation for replicate injections is not more than 2.0%.

Procedure—Separately inject equal volumes (about 25 µL) of the *Standard preparation* and the *Assay preparation* into the chromatograph, record the chromatograms, and measure the responses for the major peaks. Calculate the quantity, in mg, of $C_{18}H_{24}O_2$ in the portion of Estradiol taken by the formula:

$$5C(R_U/R_S)$$

in which C is the concentration, in µg per mL, of [USP Estradiol RS](#) in the *Standard preparation*; and R_U and R_S are the peak response ratios obtained from the *Assay preparation* and the *Standard preparation*, respectively.

Auxiliary Information - Please [check for your question in the FAQs](#) before contacting USP.

Topic/Question	Contact	Expert Committee
ESTRADIOL	Documentary Standards Support	SM52025 Small Molecules Therapeutic Areas 5
REFERENCE STANDARDS SUPPORT	RS Technical Services RSTECH@usp.org	SM52025 Small Molecules Therapeutic Areas 5

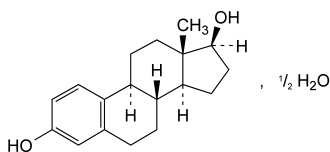
Chromatographic Database Information: [Chromatographic Database](#)

Most Recently Appeared In:
Pharmacopeial Forum: Volume No. PF 33(6)

DOI: <https://doi.org/10.31003/USPNF.M30610.02.01>

ESTRADIOL HEMIHYDRATE

Estradiolum hemihydricum



$C_{18}H_{24}O_2 \cdot \frac{1}{2}H_2O$

M_r 281.4

DEFINITION

Estra-1,3,5(10)-triene-3,17 β -diol hemihydrate.

Content: 97.0 per cent to 103.0 per cent (anhydrous substance).

CHARACTERS

Appearance: white or almost white, crystalline powder or colourless crystals.

Solubility: practically insoluble in water, soluble in acetone, sparingly soluble in ethanol (96 per cent), slightly soluble in methylene chloride.

IDENTIFICATION

First identification: B.

Second identification: A, C, D, E.

A. Melting point (2.2.14): 175 °C to 180 °C.

B. Infrared absorption spectrophotometry (2.2.24).

Comparison: estradiol hemihydrate CRS.

C. Thin-layer chromatography (2.2.27).

Test solution. Dissolve 50 mg of the substance to be examined in *methanol R* and dilute to 50 mL with the same solvent.

Reference solution (a). Dissolve 50 mg of estradiol hemihydrate CRS in *methanol R* and dilute to 50 mL with the same solvent.

Reference solution (b). Dissolve 25 mg of ethinylestradiol CRS in reference solution (a) and dilute to 25 mL with reference solution (a).

Plate: TLC silica gel plate R.

Mobile phase: ethanol (96 per cent) R, toluene R (20:80 V/V).

Application: 5 μ L.

Development: over 3/4 of the plate.

Drying: in air until the solvent has evaporated.

Detection: heat at 110 °C for 10 min. Spray the hot plate with alcoholic solution of sulfuric acid R. Heat again at 110 °C for 10 min. Allow to cool. Examine in daylight and in ultraviolet light at 365 nm.

System suitability: the chromatogram obtained with reference solution (b) shows 2 spots which may however not be completely separated.

Results: the principal spot in the chromatogram obtained with the test solution is similar in position, colour in daylight, fluorescence in ultraviolet light at 365 nm and size to the principal spot in the chromatogram obtained with reference solution (a).

D. To about 1 mg add 0.5 mL of freshly prepared sulfomolybdic reagent R2. A blue colour develops which in ultraviolet light at 365 nm has an intense green fluorescence. Add 1 mL of sulfuric acid R and 9 mL of water R. The colour becomes pink with a yellowish fluorescence.

E. Water (see Tests).

TESTS

Specific optical rotation (2.2.7): + 76.0 to + 83.0 (anhydrous substance).

Dissolve 0.250 g in ethanol (96 per cent) R and dilute to 25.0 mL with the same solvent.

Related substances. Liquid chromatography (2.2.29).

Test solution. Dissolve 25 mg of the substance to be examined in 10 mL of *acetonitrile R* and dilute to 25.0 mL with *methanol R2*.

Reference solution (a). Dilute 1.0 mL of the test solution to 100.0 mL with the mobile phase. Dilute 2.0 mL of this solution to 10.0 mL with the mobile phase.

Reference solution (b). Mix equal volumes of a 1 mg/mL solution of the substance to be examined in *methanol R2* and of a 1 mg/mL solution of 2,3-dichloro-5,6-dicyanobenzoquinone *R* in *methanol R2*. Allow to stand for 30 min before injection.

Reference solution (c). Dissolve 5 mg of *estradiol for peak identification CRS* (containing impurities A, B and C) in 2 mL of *acetonitrile R* and dilute to 5.0 mL with *methanol R2*.

Column:

- size: $l = 0.25$ m, $\varnothing = 4.6$ mm;
- stationary phase: end-capped octadecylsilyl silica gel for chromatography *R* (5 μ m).

Mobile phase: to 400 mL of *acetonitrile R* add 50 mL of *methanol R2* and 400 mL of *water for chromatography R*; allow to stand for 10 min, dilute to 1000 mL with *water for chromatography R* and mix again.

Flow rate: 1 mL/min.

Detection: spectrophotometer at 280 nm.

Injection: 20 μ L.

Run time: twice the retention time of estradiol.

Identification of impurities: use the chromatogram supplied with *estradiol for peak identification CRS* and the chromatogram obtained with reference solution (c) to identify the peaks due to impurities A, B and C; use the chromatogram obtained with reference solution (b) to identify the peak due to impurity D.

Relative retention with reference to estradiol (retention time = about 13 min): impurity D = about 0.9; impurity B = about 1.1; impurity A = about 1.4; impurity C = about 1.9.

System suitability: reference solution (c):

- resolution: minimum 2.5 between the peaks due to estradiol and impurity B.

Limits:

- correction factor: for the calculation of content, multiply the peak area of impurity D by 0.4;
- impurities A, B, C, D: for each impurity, not more than 1.5 times the area of the principal peak in the chromatogram obtained with reference solution (a) (0.3 per cent);
- unspecified impurities: for each impurity, not more than 0.5 times the area of the principal peak in the chromatogram obtained with reference solution (a) (0.10 per cent);
- total: not more than 2.5 times the area of the principal peak in the chromatogram obtained with reference solution (a) (0.5 per cent);
- disregard limit: 0.25 times the area of the principal peak in the chromatogram obtained with reference solution (a) (0.05 per cent).

Water (2.5.12): 2.9 per cent to 3.5 per cent, determined on 0.500 g.

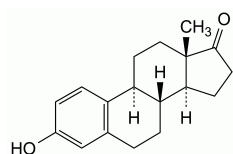
ASSAY

Dissolve 20.0 mg in *ethanol* (96 per cent) *R* and dilute to 100.0 mL with the same solvent. Dilute 5.0 mL of the solution to 50.0 mL with 0.1 M *sodium hydroxide*. Allow to cool to room temperature. Measure the absorbance (2.2.25) of the solution at the maximum at 238 nm.

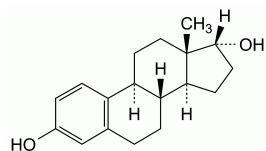
Calculate the content of $C_{18}H_{24}O_2$ taking the specific absorbance to be 335.

IMPURITIES

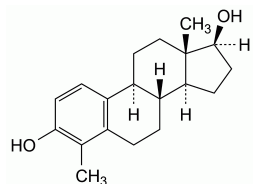
Specified impurities: A, B, C, D.



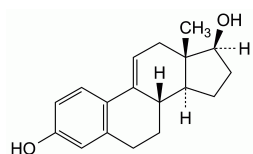
A. 3-hydroxyestra-1,3,5(10)-trien-17-one (estrone),



B. *estra-1,3,5(10)-triene-3,17α-diol (17α-estradiol),*



C. *4-methylestra-1,3,5(10)-triene-3,17β-diol,*



D. *estra-1,3,5(10),9(11)-tetraene-3,17β-diol.*